CLINICAL TRIAL FAILURES · DISCONTINUATION RATE · ONCOLOGY PHASE 2,3 · STARTS 2015–2024

HER2: 5 molecules stopped early

7 stopped trials across 5 molecules: Tarloxotinib, Trastuzumab, Trastuzumab Emtansine, Afatinib and 1 more. Against 5.2% for all oncology trials with a resolved drug target and 4.5% across all oncology Phase 2,3 trials in the same window. The rate counts trials that stopped early for efficacy, safety or benefit–risk reasons. Trials that ran to completion and missed their primary endpoint are counted separately below, never in the rate.

3.5%
7 of 202 closed trials (95% CI 1.7%–7.0%)

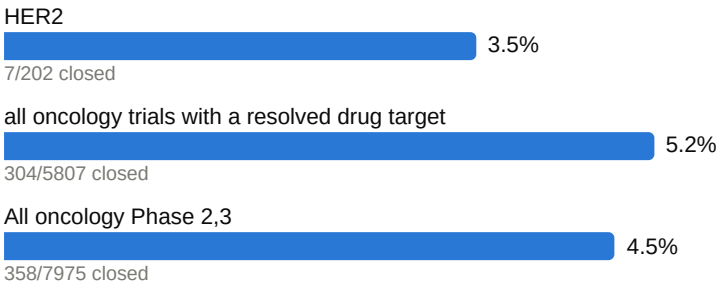
5 / 2 / 0
efficacy / safety / both (adds to 7)

26.5%
of 761 trials have closed · 559 still open or unresolved

1.3%
stopped for a biological reason within 3 years of starting (95% CI 0.6%–2.9%; 301 still at risk; 1.5% area-wide)

4 of 9 **Completed and missed the primary endpoint.** Of 144 completed trials in this segment, 9 posted a primary result on ClinicalTrials.gov that can be read — the sponsor's own comparison held to the sponsor's own threshold, or its statement that the endpoint was missed: 4 missed and 5 met. Counted separately and never in the rate above; a floor, not a rate. 60 of the 131 completed trials that finished more than a year ago have posted no results at all — a gap in what can be known, not a sign of failure.

How this compares



What is in this class: 346 distinct experimental drugs across 761 trials (small molecule 471, antibody 381, adc 268, unknown 121). Most tested: TRASTUZUMAB (220), PYROTINIB (113), PERTUZUMAB (96), DISITAMAB VEDOTIN (80), CAPECITABINE (70). A class groups drugs by what they act on; check that the grouping is one you would make before reading the rate as a property of the mechanism.

How much does this rest on one decision? The 7 stops came from 6 sponsor-asset programmes across 6 sponsors; the largest (Rain Oncology Inc) contributed 2; removing that programme's trials from both sides leaves 2.5%. Screened alongside 85 other segments, the rate is within what screening that many classes could produce (q=1); the molecules and trials that failed here are the record either way. A further 11 closed trials here were terminated with no cause recorded in the registry; if every one of them were biological the rate would be 8.9%. Read it as a screen worth checking against the underlying trials.

Bars show the share of closed trials stopped for biological reasons. Confidence intervals overlap where sample sizes are small — read the counts, not just the bars.

By start cohort

Trials started	Stops	Closed	Rate	95% CI
2015–2017	5	88	5.7%	2.5%–12.6%
2018–2020	1	73	1.4%	0.2%–7.4%
2021–2024	1	41	2.4%	0.4%–12.6%

Where the stops sit

Efficacy / futility: 5, Safety: 2.
Sponsors with most stops: Rain Oncology Inc (2), Spanish Breast Cancer Research Group (1), The Netherlands Cancer Institute (1), Washington University School of Medicine (1), University of Miami (1), Universität Duisburg-Essen (1).

The stopped trials

Trial	Ph	Sponsor	Experimental drugs	Stopped	Type	Registry stop reason
NCT05583110	2	Spanish Breast Cancer Research Group	TUCATINIB, TRASTUZUMAB, VINORELBINE	2025-05	Safety	The study was closed early due to safety concerns and an unfavorable benefit-risk balance. Unexpected high neutropenia rates required a costly sub-study, which was not feasible, forcing the sponsor to terminate the trial prematurely.

Trial	Ph	Sponsor	Experimental drugs	Stopped	Type	Registry stop reason
NCT03784599	2	The Netherlands Cancer Institute	TRASTUZUMAB EMTANSINE, OSIMERTINIB	2021-05	Efficacy	insufficient effectiveness
NCT02907918	2	Washington University School of Medicine	PALBOCICLIB, LETROZOLE, TRASTUZUMAB, GOSERELIN	2020-08	Efficacy	Futility
NCT02658084	2	University of Miami	VINORELBINE, TRASTUZUMAB EMTANSINE	2018-10	Safety	Terminated due to low accrual and toxicity concerns.
NCT02595840	2	Universität Duisburg-Essen	AFATINIB, PEMETREXED	2017-11	Efficacy	Futility
NCT02454842	2	Rain Oncology Inc	TARLOXOTINIB	2017-01	Efficacy	Lack of efficacy

The 6 most recent of 7 stopped trials. Every molecule behind all 7 is named above; the complete trial list, the trials still running and the terminations with no readable cause ship with the dataset and the diligence report.

Method
Denominator: ClinicalTrials.gov interventional Phase 2,3 oncology trials started 2015–2024 that have closed (completed or terminated). Numerator: terminated trials whose registry stop reason is classified as biological (efficacy, safety or benefit–risk) by Classification V2 — held-out precision 95.5%, recall 95.3% (n=600). Drugs are linked to ChEMBL and the NCI Thesaurus; 77% of industry oncology trials in this window carry a resolved drug target, and a trial without one cannot enter a mechanism class. Intervals are Wilson 95%.

Limits
Not a failure rate: completed trials that missed their endpoints are not in the rate. They are counted separately where the sponsor posted a primary result that can be read — most did not, so that count is a floor — and programmes discontinued after a completed trial do not appear. Stop reasons are sponsor-reported. This is a closed-trial proportion, not a time-to-event analysis: only 26.5% of trials in this segment have closed, and a trial that stops early enters the denominator sooner than one that runs to completion, which can inflate the rate in immature segments. Recent cohorts have fewer closed trials, so their rates are less stable. Research signals, not clinical or investment advice.

The whole cohort, trial by trial. The diligence report names every trial behind these figures — the ones still running and the ones that completed and missed their primary endpoint included — and places your molecule against each one that failed. Free during the beta: enter your work email at **clinicaltrialfailures.com/asset-check** and it opens straight away.

Sources: ClinicalTrials.gov (NLM); ChEMBL (EMBL-EBI, CC BY-SA 3.0); NCI Thesaurus (NCI); RxNorm/RxClass (NLM); SEC EDGAR. Classification, linkage and rates by Clinical Trial Failures. Rebuilt weekly; this brief covers trials started 2015–2024.